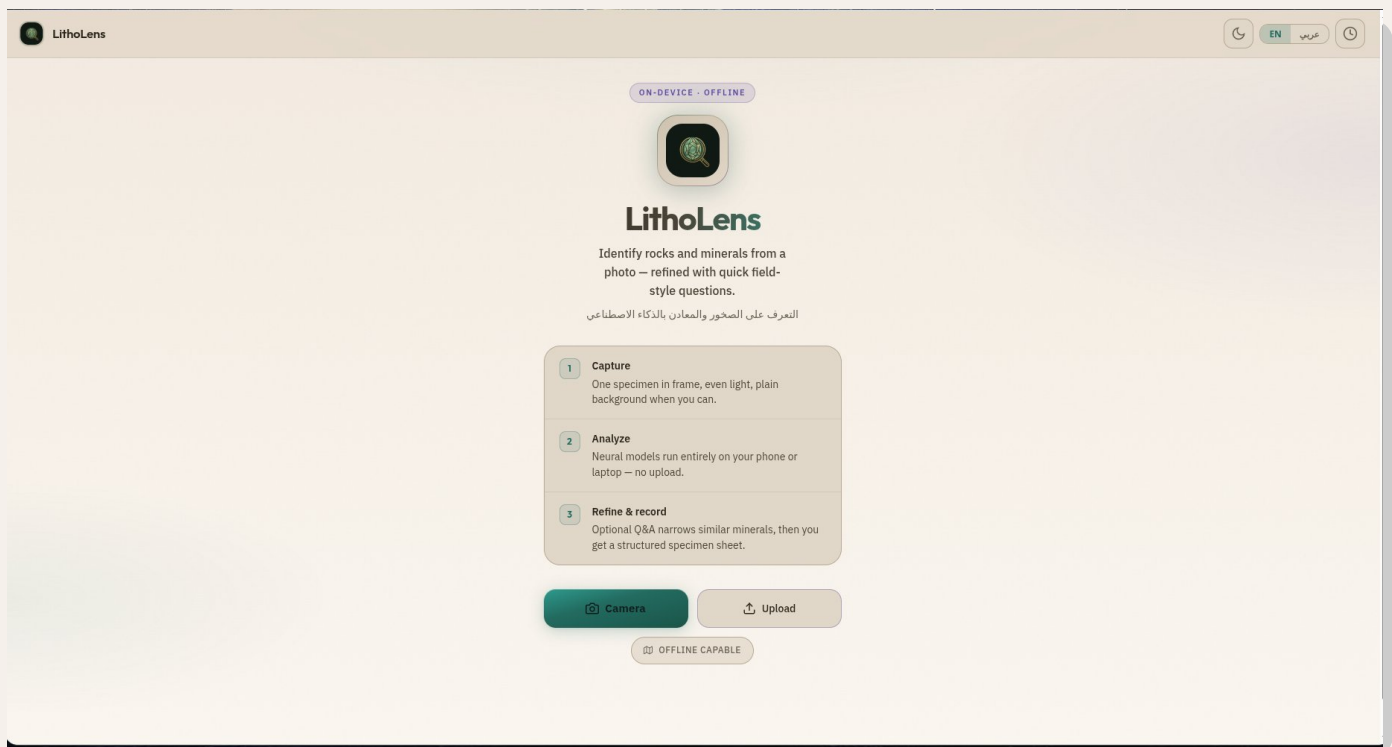




LithoLens

Rock & Mineral Identification
Powered by On-Device AI

Browser-based · Offline-capable · English / Arabic



Home Screen — Capture or Upload

The app opens with a clean landing screen. Users choose between Camera (live capture) or Upload (gallery file). The app runs entirely on-device — no internet needed after the first load.

Crop the specimen

Drag to select only the mineral specimen — exclude hands and background.



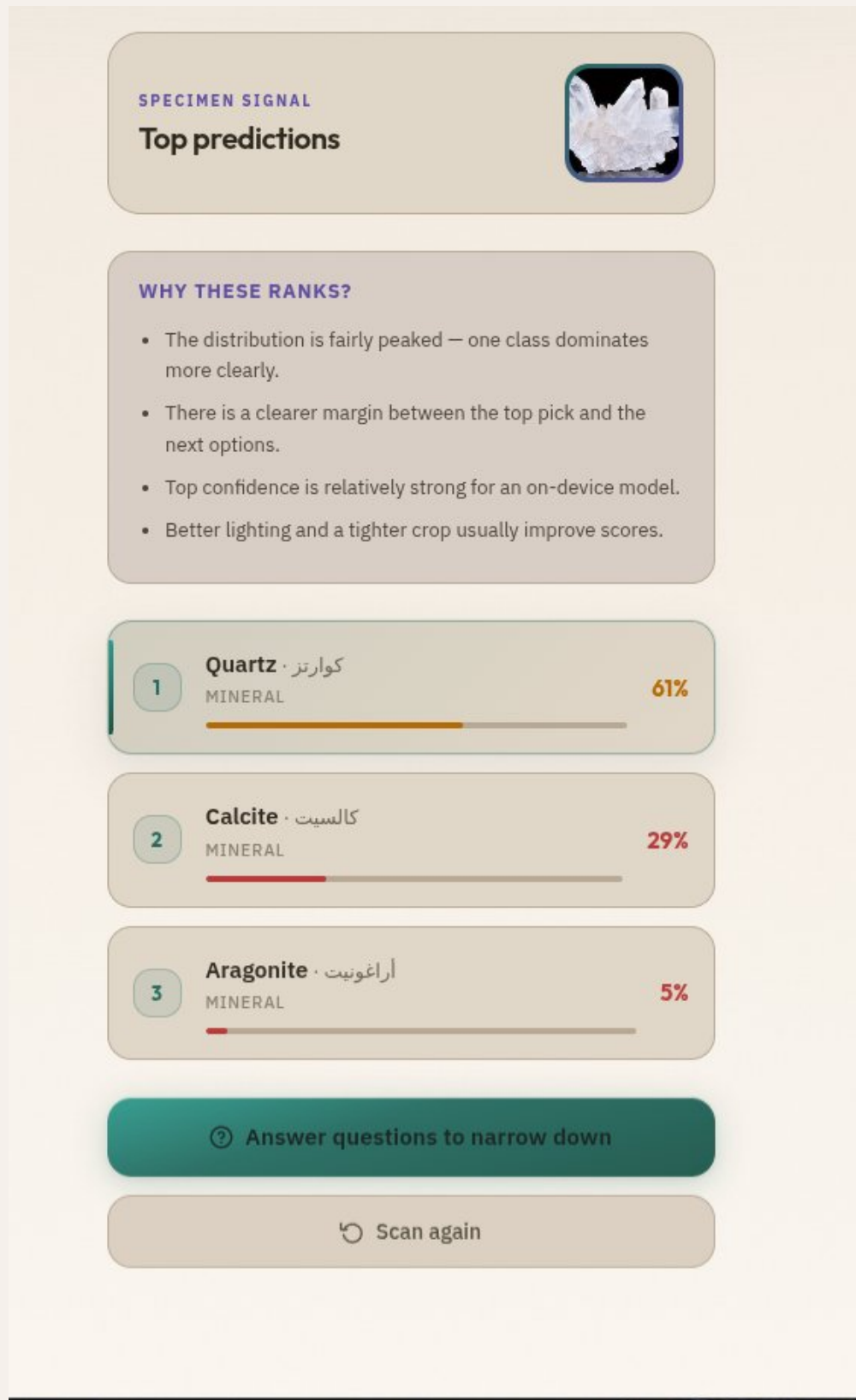
✓ Analyze cropped region

☒ Skip — use full image

🔄 Retake

Crop the Specimen

Users can drag to crop the image and isolate the mineral specimen, excluding hands and background for better accuracy before analysis.



AI Predictions with Confidence Scores

The custom ONNX classifier returns ranked predictions with confidence percentages. Here: Quartz (61%), Calcite (29%), Aragonite (5%). Includes an explanation of why these ranks were produced.

STEP 1

Let's narrow it down

Comparing: Quartz vs Calcite

Put a drop of vinegar or acid on the surface. What happens?

Nothing happens

Slight fizzing

Strong fizzing / bubbling

Q&A Disambiguation

When top predictions are close, the app asks field-style questions (e.g. acid test) to narrow down candidates. This uses real mineralogical properties from the local minerals database — no internet required.

✓ IDENTIFIED



Quartz
كوارتز

MINERAL

⚡ HARDNESS

7 Mohs

☆ LUSTER

vitreous

○ STREAK

white

||| CLEAVAGE

none

🧪 ACID TEST

No

هذا معدن الكوارتز، صلابته 7، لمعانه زجاجي، مخدشه أبيض،
HCl ليس له انفصام، ومكسره محاري، لا يتفاعل مع حمض.

Download card (PNG)

Share card

Print / Save as PDF

🔄 Identify another sample

Identified — Specimen Card

The final identified specimen sheet shows key properties: hardness, luster, streak, cleavage, and acid test. Available in English and Arabic. Users can Download as PNG, Share, or Print/Save as PDF.